In 1988, Lance Dixon of the Stanford Linear Accelerator Center made a pivotal observation in this regard that was further amplified by Wolfgang Lerche of CERN, Vafa at Harvard, and Nicholas Warner, then of the Massachusetts Institute of Technology. Based upon aesthetic arguments rooted in considerations of symmetry, these physicists made the audacious suggestion that it might be possible for two different Calabi-Yau shapes, chosen for the extra curled-up dimensions in string theory, to give rise to identical physics.

To give you an idea of how this rather far-fetched possibility might actually occur, recall that the number of holes in the extra Calabi-Yau dimensions determines the number of families into which string excitations will arrange themselves. These holes are analogous to the holes one finds in a torus or its multihandled cousins, as illustrated in Figure 9.1. One deficiency of the two-dimensional figure that we must show on the printed page is that it cannot show that a six-dimensional Calabi-Yau space can have holes of a variety of dimensions. Although such holes are harder to picture, they can be described with well-understood mathematics. A key fact is that the number of families of particles arising from string vibrations is sensitive only to the total number of holes, not to the number of holes of each particular dimension (that's why, for instance, we did not worry about drawing distinctions between the different types of holes in our discussion in Chapter 9). Imagine, then, two Calabi-Yau spaces in which the number of holes in various dimensions differs, but in which the total number of holes is the same. Since the number of holes in each dimension is not the same, the two Calabi-Yaus have different shapes. But since they have the same total number of holes, each yields a universe with the same number of families. This, of course, is but one physical property. Agreement on all physical properties is a far more restrictive requirement, but this at least gives the flavor of how the Dixon-Lerche-Vafa-Warner conjecture could possibly be true.

In the fall of 1987, I joined the physics department at Harvard as a postdoctoral fellow and my office was just down the hall from Vafa's. As my thesis research had focused on the physical and mathematical properties of curled-up Calabi-Yau dimensions in string theory, Vafa kept me closely apprised of his work in this area. When he stopped by my office in the fall of 1988 and told me of the conjecture that he,